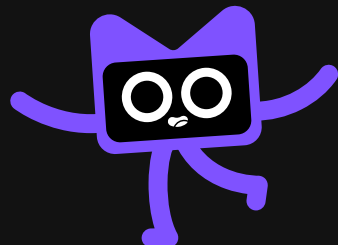


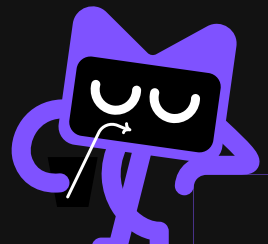
Building Production-Ready Kubernetes Operators: A Practical Guide

Frederik Pietzko

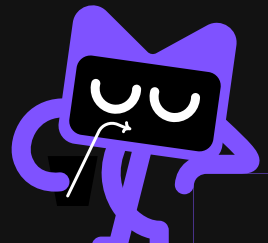
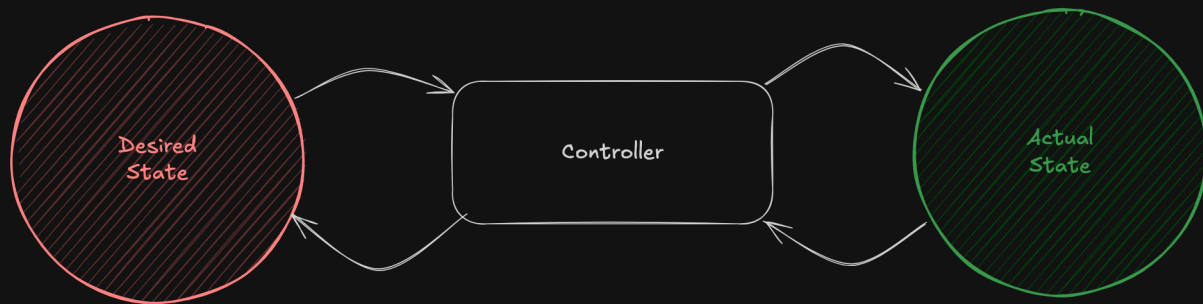


What is a Kubernetes Operator?

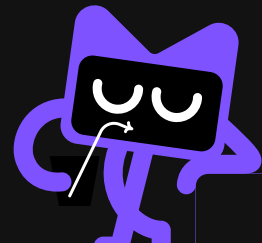
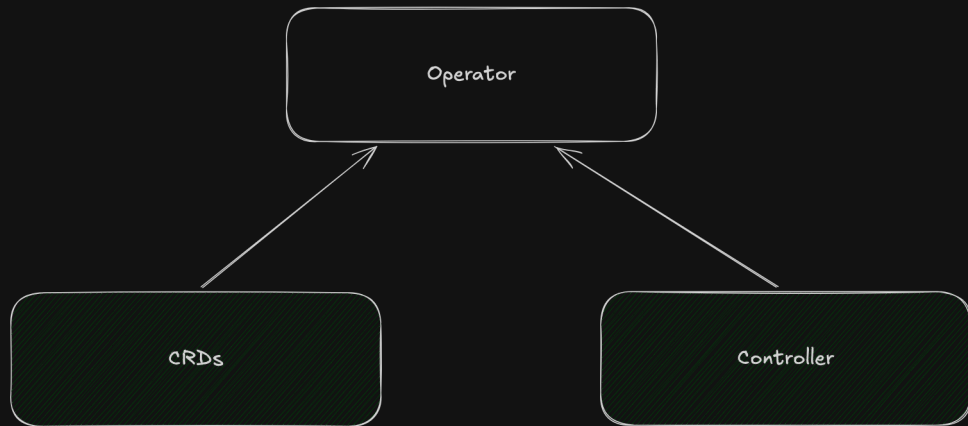
- Encode **domain expertise** as custom controllers
- e.g.: a DB Operator for failovers, backups, migrations...
- Operators you might know:
 - ArgoCD
 - cert-manager
 - Keycloak Operator
 - Crossplane



The Reconciliation Loop



What are Operators made of?



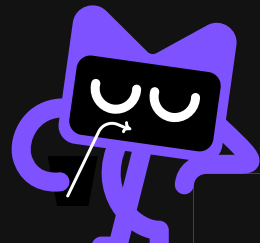
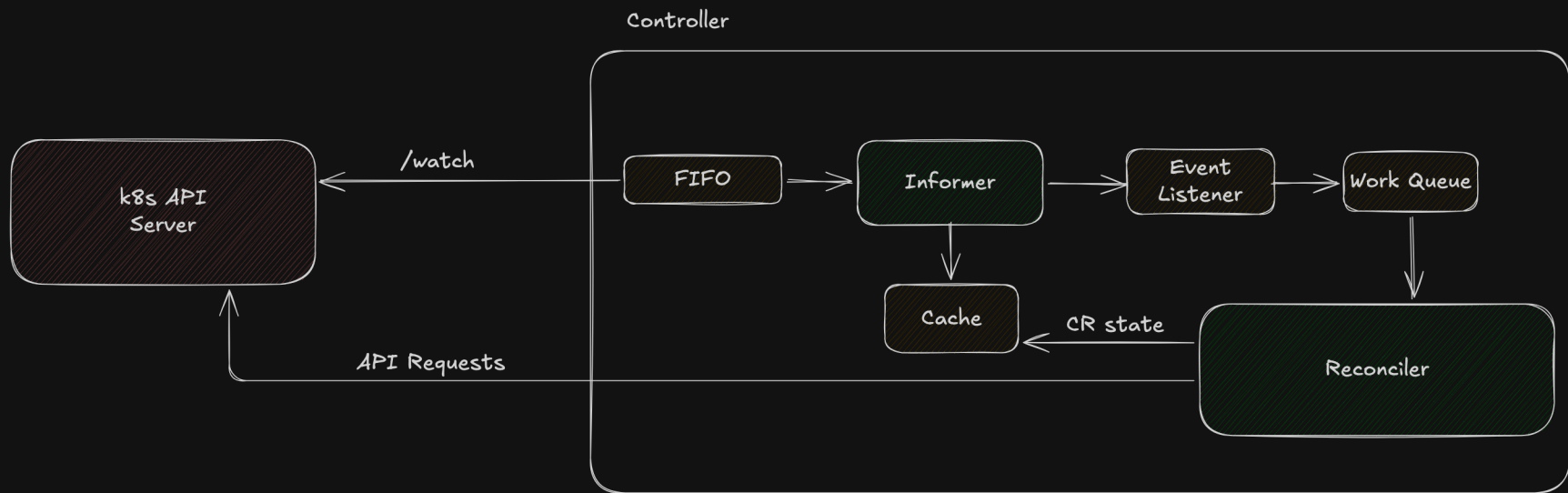
Custom Resource Definitions

Describe the API of the operator

```
apiVersion: "apiextensions.k8s.io/v1"
kind: "CustomResourceDefinition"
spec:
  ...
  versions:
    - name: "v1"
      schema:
        openAPIV3Schema:
          properties:
            spec:
              properties:
                imageName:
                  type: "string"
              type: "object"
```



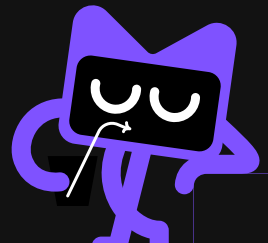
How do Controllers work?



Reconciliation

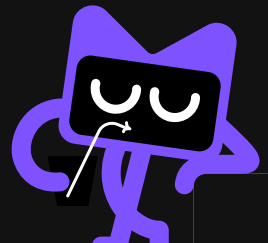
The whole thing in action

- Build Small Demo
- Operator that dispatches Code Review Agents in a Job



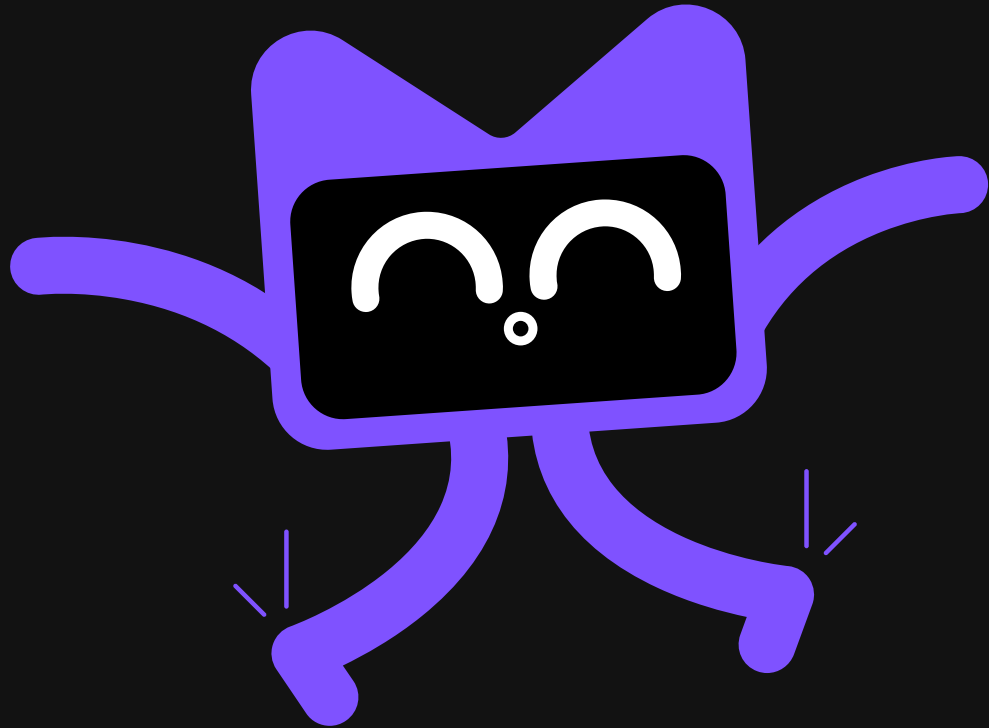
Target Flow

- Apply custom resource `AgentReviewRequest` to the default namespace
- Operator creates ConfigMap with Repo & PR Details
- Operator spawns Agent Job and mounts ConfigMap
- Agent Job create custom resource `ReviewResult` with comments
- Operator continuously updates status on `AgentReviewRequest`



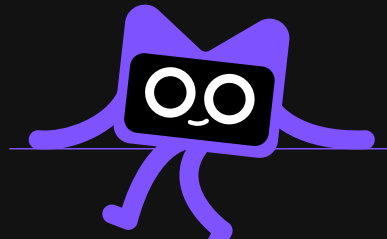
Demo

Let's write some code!



Deploying the Operator

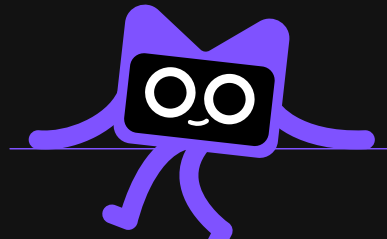
- Apply CRDs to cluster
- setup RBAC & Service Account
- Deploy operator as a normal Deployment with a single replica



But what about high availability?

- Scale horizontally!
- **BUT:** we don't want multiple controllers reconciling the same resource
- Operators do leader election

```
@Bean
fun operatorConfiguration(): Consumer<ConfigurationServiceOverride> = Consumer { override ->
    val leaderConfig =
        LeaderElectionConfigurationBuilder.aLeaderElectionConfiguration("review-agent-operator")
            .withLeaseNamespace("default")
            .build()
    override.withLeaderElectionConfiguration(leaderConfig)
}
```



Thank you for listening!

- Time for Q&A



Code & Slides

